

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AP5017
Subject Title	Advanced Materials Analysis and Characterization
Credit Value	3
Level	5
Pre-requisite / Co-requisite/ Exclusion	None
Objectives	To introduce knowledge in advanced instrumentation for materials analysis to students.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a) have an understanding of the principles and applications of a selected range of advanced instruments for materials analysis, b) understand the methodology of materials testing for quality assurance and failure analysis, and c) develop students' experimental skills through laboratory work experience.
Subject Synopsis/ Indicative Syllabus	• Overview of the principles and techniques in materials characterization and failure analysis; testing codes and standards. • Non-destructive testing methods: dye penetration, magnetic particles inspection, eddy currents, ultrasonic and radiography. • Mechanical and thermal techniques: dynamic mechanical analysis, thermomechanical analysis, and differential scanning calorimetry. • Microscopy: TEM, SEM, AFM, SAM and SLAM. • Other structural, chemical and surface analyses: XRD, FTIR, Raman spectroscopy, RHEED, RBS, EDX, LIMS and other novel techniques using plasma and post-ionization. • Hands-on experiments using some of the available advanced instruments/facilities in our research centers.
Teaching/Learning Methodology	In order to stimulate and motivate the students' interest in the study of materials science and related topics, hands-on experiments will be offered for students to gain experience in the characterization of the electrical and physical properties of some materials.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)		
			a	b	c
	1. Examination	60	✓	✓	✓
	2. Continuous assessments	40	✓	✓	✓
	Total	100 %			
Continuous assessments will strengthen the students' basic knowledge and the analytical skill to solve the problems related to different advanced measurement techniques for materials. Tests will review their understanding of the course. The examination will accelerate their knowledge's understanding and improve their manipulation of problem-solving. Hence, the proposed assessment methods are necessary to assess the intended learning outcomes.					
Student Study Effort Expected	Class contact:				
	▪ Lectures/Seminar				27 Hrs.
	▪ Laboratory				12 Hrs.
	Other student study efforts:				
	▪ Self-study				81 Hrs.
	The total student study effort				120 Hrs.
Reading List and References	• C. Hellier,' Handbook of Nondestructive Evaluation', McGraw-Hill, 3rd Edition, 2020. • P.J. Shull (Ed.),' Nondestructive Evaluation', Marcel Dekker, 2002. • Frank H. Chung and Deane K. Smith,' Industrial Applications of X-ray Diffraction', Marcel Dekker, eBook, 2014. • J.I. Goldstein,' Scanning Electron Microscopy and X-Ray Microanalysis: A Text for Biologists, Materials Scientists, and Geologists', 2nd Edition, Kluwer Academic, Publishers, 1992. • • J.I. Goldstein, <i>etal</i>, 'Scanning Electron Microscopy and X-Ray Microanalysis 4th edition, Springer, 2018.				